

"Network-1 Project Report"

The Secure Financial Multi-Branch Banking Hub :

1. Project Overview

The Secure Financial Multi-Branch Banking Hub is a high-security banking network designed using Cisco Packet Tracer. The project simulates communication between a Headquarter (HQ), two branch offices, and a Central Bank router through a WAN infrastructure.

The network was designed to provide secure and reliable communication between all branches while implementing important networking concepts such as VLANs, VLSM subnetting, DHCP, DNS, NAT, PAT, Static Routing, Web Server hosting, and IPsec VPN tunnels.

Each branch network is divided into separate VLANs for ATM devices and Teller departments to improve security and network management. DHCP is used to assign IP addresses dynamically to clients, while DNS resolves the banking domain name to the internal web server.

Static routing ensures stable paths between routers, and VPN tunnels secure communication between the banking branches over the WAN.

2. Network Topology

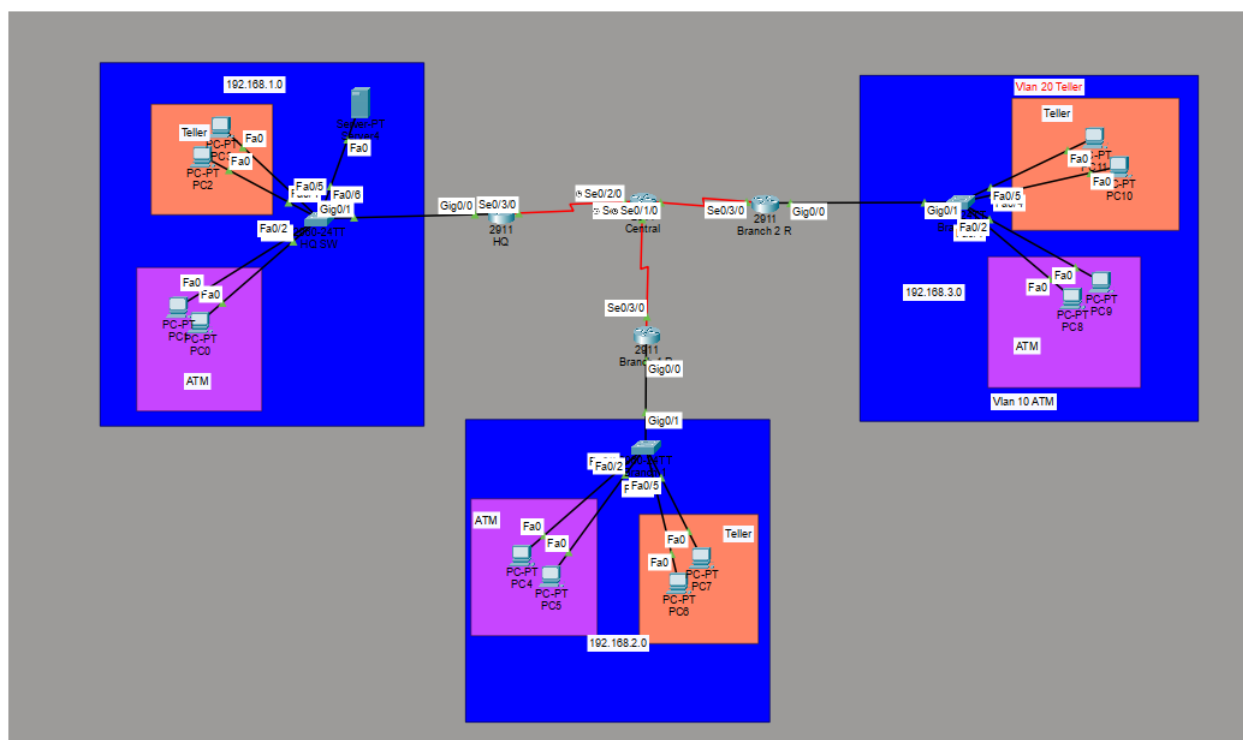
The network topology consists of:

- 1 HQ Router
- 1 Central Bank Router
- 2 Branch Routers
- 3 Switches

- ATM VLAN
- Teller VLAN
- DNS Server
- Internal Web Server
- WAN Serial Connections

Network Devices

Device	Purpose
HQ Router	Connects HQ network to WAN
Central Router	Main WAN communication router
Branch Routers	Connect branch LANs to WAN
Switches	VLAN segmentation
DNS Server	Resolves banking domain
Web Server	Hosts banking login page
PCs	ATM and Teller clients



3. VLSM Calculation

Variable Length Subnet Masking (VLSM) was used to efficiently divide the network into multiple subnets based on the required number of hosts.

The ATM VLAN requires fewer hosts than the Teller VLAN, therefore different subnet sizes were assigned to reduce IP address waste.

VLSM Table

Network	Required Hosts	Subnet Mask	CIDR
HQ LAN	50 Hosts	255.255.255.192	/26
Branch 1 LAN	50 Hosts	255.255.255.192	/26
Branch 2 LAN	50 Hosts	255.255.255.192	/26
WAN Links	2 Hosts	255.255.255.252	/30

4. Addressing Table

LAN Addressing

Network	Network ID	First IP	Last IP	Broadcast
HQ Network	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63
Branch 1 Network	192.168.2.0/26	192.168.2.1	192.168.2.62	192.168.2.63
Branch 2 Network	192.168.3.0/26	192.168.3.1	192.168.3.62	192.168.3.63

WAN Addressing

WAN Link	Network ID	First IP	Last IP	Broadcast
HQ ↔ Central	10.0.0.0/30	10.0.0.1	10.0.0.2	10.0.0.3
Central ↔ Branch 1	10.0.0.4/30	10.0.0.5	10.0.0.6	10.0.0.7
Central ↔ Branch 2	10.0.0.8/30	10.0.0.9	10.0.0.10	10.0.0.11

Verification Commands

show ip interface brief

Screenshot Section

HQ Router

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES NVRAM    up          up
GigabitEthernet0/0.10 192.168.1.65   YES NVRAM    up          up
GigabitEthernet0/0.20 192.168.1.1    YES NVRAM    up          up
GigabitEthernet0/1    unassigned      YES NVRAM    administratively down down
GigabitEthernet0/2    unassigned      YES NVRAM    administratively down down
Serial10/3/0         10.0.0.1       YES NVRAM    up          up
Serial10/3/1         unassigned      YES NVRAM    administratively down down
Tunnel0              172.16.1.1     YES NVRAM    up          up
Vlan1                unassigned      YES NVRAM    administratively down down
_
```

Branch 1 Router

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	NVRAM	up	up
GigabitEthernet0/0.10	192.168.2.65	YES	NVRAM	up	up
GigabitEthernet0/0.20	192.168.2.1	YES	NVRAM	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/3/0	10.0.0.5	YES	NVRAM	up	up
Serial0/3/1	unassigned	YES	NVRAM	administratively down	down
Tunnel1	172.16.2.1	YES	NVRAM	up	up
Vlan1	unassigned	YES	NVRAM	administratively down	down

Branch 2 Router

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	NVRAM	up	up
GigabitEthernet0/0.10	192.168.3.65	YES	NVRAM	up	up
GigabitEthernet0/0.20	192.168.3.1	YES	NVRAM	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/3/0	10.0.0.9	YES	NVRAM	up	up
Serial0/3/1	unassigned	YES	NVRAM	administratively down	down
Tunnel2	172.16.3.1	YES	NVRAM	up	up
Vlan1	unassigned	YES	NVRAM	administratively down	down

5. VLAN Configuration

VLANs were created to separate ATM devices from Teller devices in every branch. VLAN segmentation improves network organization, security, and traffic management.

VLAN Structure

VLAN ID	VLAN Name	Department
---------	-----------	------------

10	ATM	ATM Devices
----	-----	-------------

20	Teller	Teller Department
----	--------	-------------------

Switch Configuration Example

enable

configure terminal

vlan 10

name ATM

vlan 20

name Teller

interface range fa0/1-2

switchport mode access

switchport access vlan 10

interface range fa0/5-6

switchport mode access

switchport access vlan 20

Verification Commands

show vlan brief

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	ATM	active	Fa0/1, Fa0/2, Fa0/3
20	Teller	active	Fa0/4, Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

show interfaces trunk

```
Switch#show interfaces trunk
Port      Mode           Encapsulation  Status        Native vlan
Gig0/1    on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20
```

6. DHCP Configuration

DHCP was configured on routers to automatically assign IP addresses to Teller and ATM PCs.

This configuration reduces manual IP assignment errors and simplifies network management.

DHCP Configuration Example

```
ip dhcp pool TELLER
```

```
network 192.168.1.0 255.255.255.192
```

```
default-router 192.168.1.1
```

```
dns-server 192.168.1.10
```

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

Verification Commands

```
show ip dhcp pool
```

Screenshot Section

```
Router#show ip dhcp pool
```

```
Pool TELLER_HQ :
```

```
Utilization mark (high/low) : 100 / 0  
Subnet size (first/next)    : 0 / 0  
Total addresses             : 62  
Leased addresses            : 0  
Excluded addresses          : 1  
Pending event               : none
```

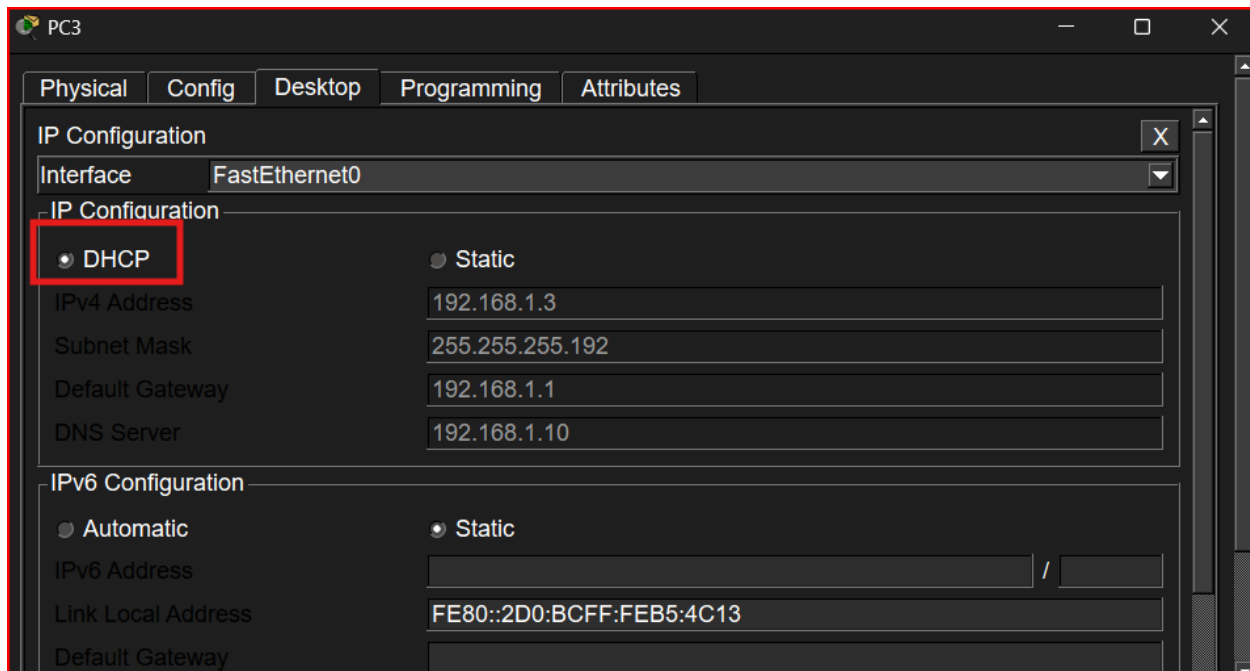
```
1 subnet is currently in the pool
```

Current index	IP address range	Leased/Excluded/Total
192.168.1.1	192.168.1.1 - 192.168.1.62	0 / 1 / 62

```
Pool TELLER :
```

```
Utilization mark (high/low) : 100 / 0  
Subnet size (first/next)    : 0 / 0  
Total addresses             : 511  
Leased addresses            : 0  
Excluded addresses          : 1  
Pending event               : none
```

```
0 subnet is currently in the pool
```



7. DNS and Web Server Configuration

A DNS server was configured to resolve the banking domain name to the internal web server IP address.

The web server hosts a simple banking login page that can be accessed using the configured domain name.

DNS Configuration

Domain Name IP Address

login.bank Web Server IP

Web Server Features

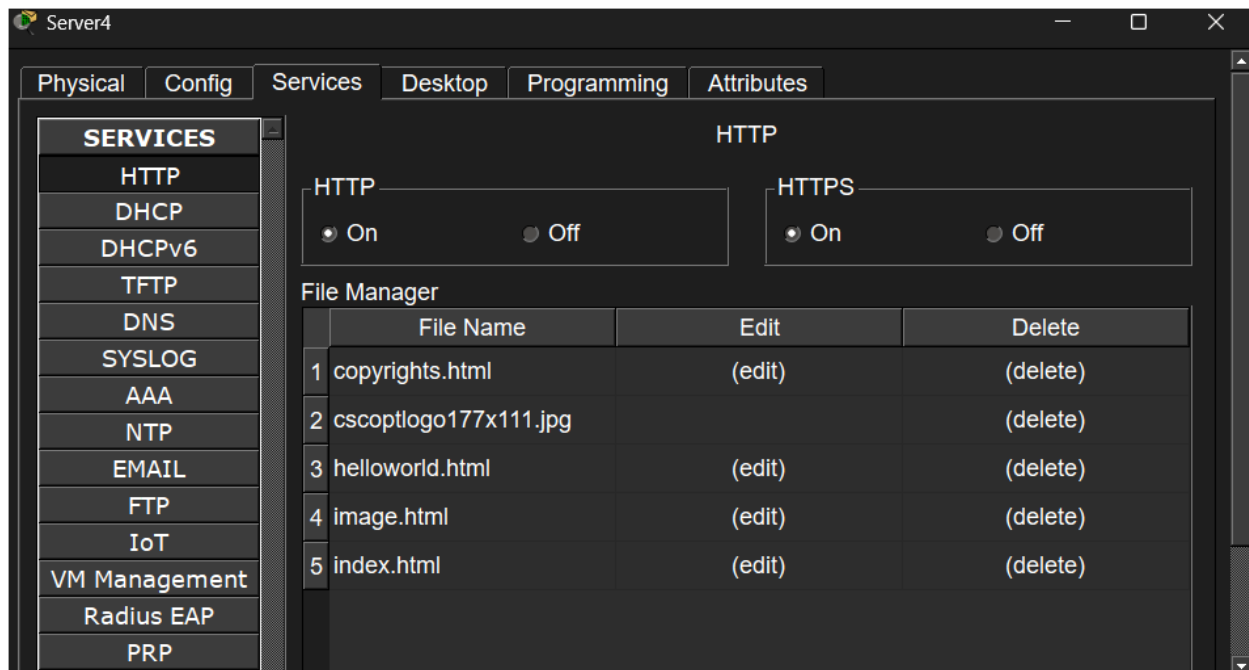
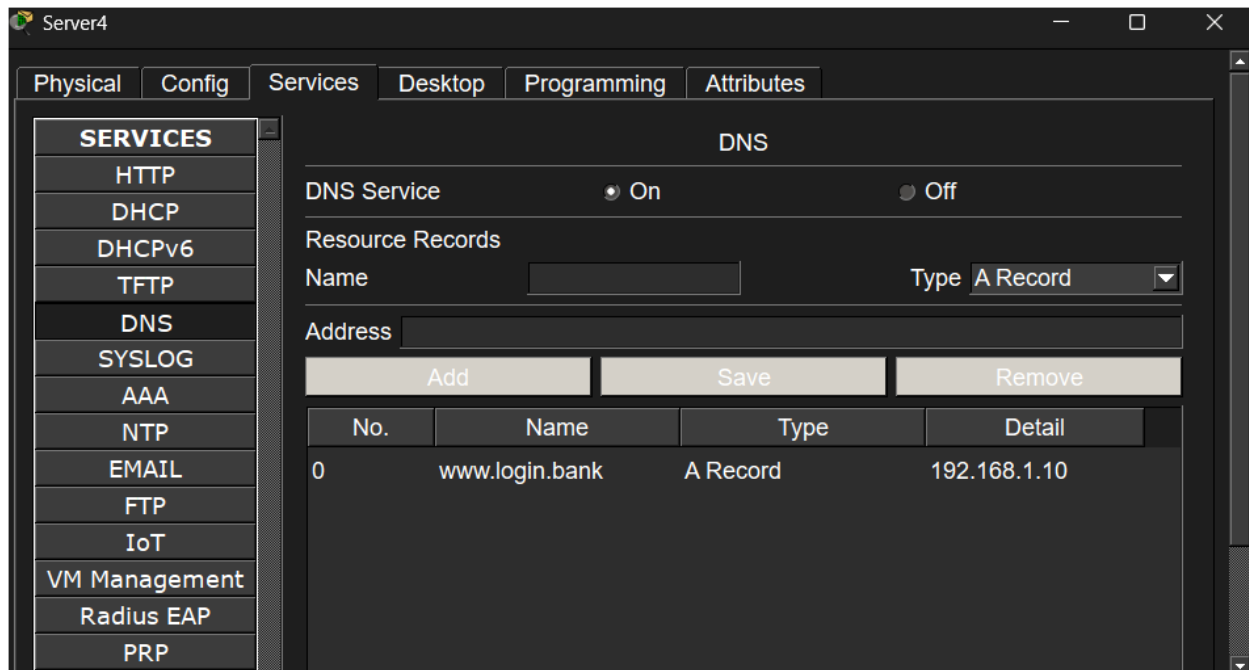
- Banking Login Page
- Internal Banking Access
- Domain-Based Access Using DNS

Verification

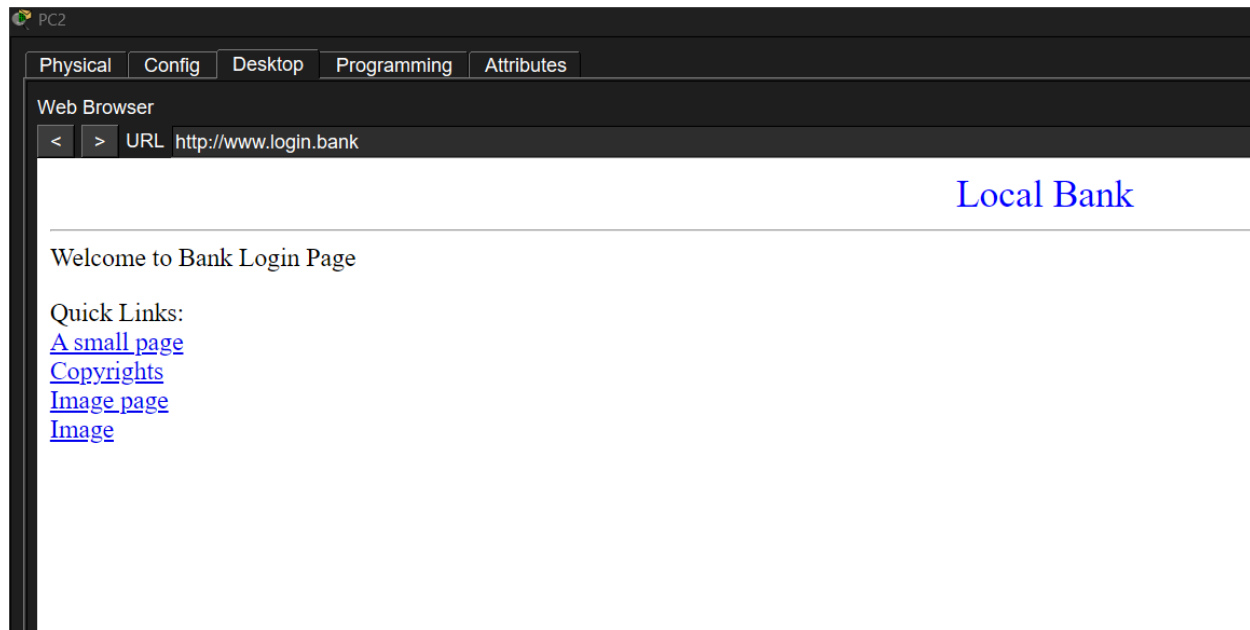
The website was successfully opened from Branch 1 and Branch 2 using the banking domain name.

Screenshot Section

DNS IP ->192.168.1.10 -> www.login.bank



From a Teller PC



8. Static Routing Configuration

Static routing was used between all routers to ensure reliable communication across the WAN.

Static routes provide fixed paths between banking branches and improve routing stability.

Static Route Configuration Example

```
ip route 192.168.2.0 255.255.255.192 10.0.0.2
```

```
ip route 192.168.3.0 255.255.255.192 10.0.0.6
```

Verification Commands

```
show ip route
```

Screenshot Section

HQ

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
S       10.0.0.4/30 [1/0] via 10.0.0.2
S       10.0.0.8/30 [1/0] via 10.0.0.2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, Tunnel0
L       172.16.1.1/32 is directly connected, Tunnel0
      192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.20
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.20
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.10
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.10
S       192.168.2.0/24 is directly connected, Serial0/3/0
S       192.168.3.0/24 is directly connected, Serial0/3/0
S       192.168.4.0/24 [1/0] via 172.16.1.2

```

Router#

Branch 1

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.0.0.0/30 [1/0] via 10.0.0.6
C       10.0.0.4/30 is directly connected, Serial0/3/0
L       10.0.0.5/32 is directly connected, Serial0/3/0
S       192.168.1.0/24 [1/0] via 10.0.0.6
      192.168.2.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.2.0/26 is directly connected, GigabitEthernet0/0.20
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0.20
C       192.168.2.64/28 is directly connected, GigabitEthernet0/0.10
L       192.168.2.65/32 is directly connected, GigabitEthernet0/0.10
S       192.168.3.0/24 [1/0] via 10.0.0.6

```

Branch 2

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.0.0.0/30 [1/0] via 10.0.0.10
C       10.0.0.8/30 is directly connected, Serial0/3/0
L       10.0.0.9/32 is directly connected, Serial0/3/0
S       192.168.1.0/24 [1/0] via 10.0.0.10
S       192.168.2.0/24 [1/0] via 10.0.0.10
      192.168.3.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.3.0/26 is directly connected, GigabitEthernet0/0.20
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0.20
C       192.168.3.64/28 is directly connected, GigabitEthernet0/0.10
L       192.168.3.65/32 is directly connected, GigabitEthernet0/0.10

```

Route Type	Destination Network	Subnet Mask	Next Hop / Exit Interface	Description
C	10.0.0.0	255.255.255.252	Serial0/3/0	WAN link between HQ and Central Router
C	172.16.1.0	255.255.255.252	Tunnel0	Tunnel link between HQ and Central Router
C	192.168.1.0	255.255.255.192	GigabitEthernet0/0.20	HQ Teller VLAN
C	192.168.1.64	255.255.255.240	GigabitEthernet0/0.10	HQ ATM VLAN
S	10.0.0.4	255.255.255.252	10.0.0.2	Route to Branch 1 WAN network
S	10.0.0.8	255.255.255.252	10.0.0.2	Route to Branch 2 WAN network
S	192.168.2.0	255.255.255.0	Serial0/3/0	Route to Branch 1 LAN
S	192.168.3.0	255.255.255.0	Serial0/3/0	Route to Branch 2 LAN
S	192.168.4.0	255.255.255.0	172.16.1.2	Route to Central Bank network

9. NAT and PAT Configuration

Network Address Translation (NAT) was implemented to hide internal private IP addresses and allow communication through public addresses.

PAT (Port Address Translation) was configured for internal users to share a single public IP address.

Static NAT was configured for the web server to allow external access.

PAT Configuration Example

```
access-list 1 permit 192.168.0.0 0.0.255.255
```

```
ip nat inside source list 1 interface serial0/0/0 overload
```

Static NAT Configuration Example

```
ip nat inside source static 192.168.1.10 10.0.0.1
```

Verification Commands

```
show ip nat translations
```

```
show ip nat statistics
```

Screenshot Section

HQ

```
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 10.0.0.1:80        192.168.1.10:80  ---               ---
```

```
Router#show ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 1 extended)
Outside Interfaces: Serial0/3/0
Inside Interfaces: GigabitEthernet0/0.10 , GigabitEthernet0/0.20
Hits: 0 Misses: 574
Expired translations: 0
Dynamic mappings:
```

Branch 2 ping from pc to WAN 10.0.0.9

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local     Outside global
icmp 10.0.0.9:5         192.168.3.2:5    10.0.0.10:5      10.0.0.10:5
icmp 10.0.0.9:6         192.168.3.2:6    10.0.0.10:6      10.0.0.10:6
icmp 10.0.0.9:7         192.168.3.2:7    10.0.0.10:7      10.0.0.10:7
icmp 10.0.0.9:8         192.168.3.2:8    10.0.0.10:8      10.0.0.10:8
Router#
```

10. Static NAT for Web Server

Static NAT was configured to allow external branches to access the internal banking web server using a public IP address.

This configuration permanently maps the internal web server IP address to a public IP address.

Static NAT Configuration

```
ip nat inside source static 192.168.1.10 10.0.0.1
```

Verification

The web server was successfully accessed from Branch 1 and Branch 2 using:

http://10.0.0.1

Verification Commands

```
show ip nat translations
```

```
show ip nat statistics
```

Screenshot Section

The screenshot shows a PC11 interface with a web browser and a terminal window.

Web Browser:

- Physical | Config | Desktop | Programming | Attributes
- Web Browser
- URL: http://10.0.0.1
- Local Bank
- Welcome to Bank Login Page
- Quick Links:
 - [A small page](#)
 - [Copyrights](#)
 - [Image page](#)
 - [Image](#)

Terminal Window:

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  10.0.0.1:80          192.168.1.10:80   ---               ---
tcp  10.0.0.1:80          192.168.1.10:80   10.0.0.5:1031     10.0.0.5:1031
tcp  10.0.0.1:80          192.168.1.10:80   10.0.0.9:1025     10.0.0.9:1025
tcp  10.0.0.1:80          192.168.1.10:80   10.0.0.9:1050     10.0.0.9:1050
tcp  10.0.0.1:80          192.168.1.10:80   10.0.0.9:1051     10.0.0.9:1051
```

11. IPsec VPN Configuration

Site-to-Site IPsec VPN tunnels were configured between the banking branches and the central router.

The VPN tunnels secure communication across the WAN by encrypting transmitted data.

VPN Features

- Encrypted Communication
- Secure Inter-Branch Connectivity
- Protected Financial Transactions

Verification Commands

show crypto isakmp sa

show crypto ipsec sa

VPN Status

The VPN tunnel status successfully reached:

QM_IDLE

which confirms that the VPN tunnel is active and operational.

Screenshot Section

```
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.2     10.0.0.1     QM_IDLE        1071      0  ACTIVE

IPv6 Crypto ISAKMP SA
```

```
Router#show crypto ipsec sa
```

```
interface: Serial10/3/0
  Crypto map tag: BANK-MAP, local addr 10.0.0.9

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  current_peer 10.0.0.10 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 1, #pkts encrypt: 1, #pkts digest: 0
    #pkts decaps: 24, #pkts decrypt: 24, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.0.0.9, remote crypto endpt.:10.0.0.10
    path mtu 1500, ip mtu 1500, ip mtu idb Serial10/3/0
    current outbound spi: 0xB93097FE(3106969598)

  inbound esp sas:
```

12. ICMP Connectivity Testing

ICMP testing was performed to verify communication between all branches and network devices.

Ping tests confirmed successful communication through the WAN and VPN tunnels.

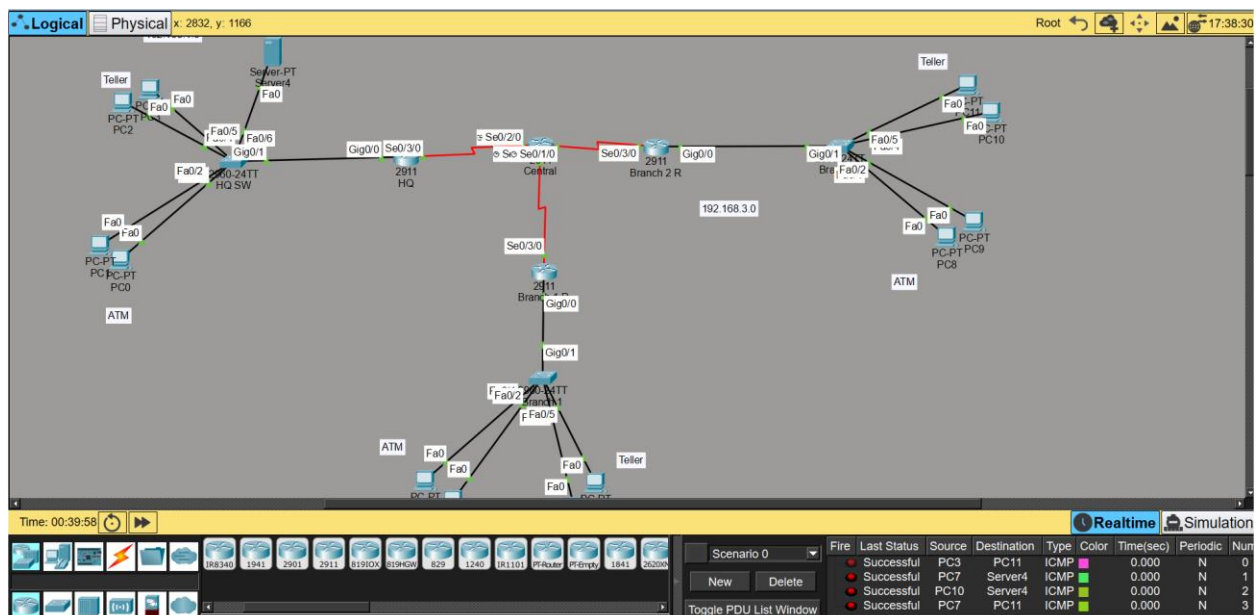
Test Results

Source	Destination	Result
HQ	Branch 1	Successful
HQ	Branch 2	Successful
Branch 1	Web Server	Successful
Branch 2	Web Server	Successful
Branch 1	Branch 2	Successful

Verification Command

ping [destination IP]

Screenshot Section



13. Security Features

Several security techniques were implemented in the project to protect banking communication and services.

Implemented Security Features

- VLAN Segmentation
- IPsec VPN Encryption
- NAT/PAT Address Hiding
- Internal DNS Resolution
- Controlled WAN Communication
- Static Routing Stability

These features improve network confidentiality, organization, and secure communication between banking branches.

14. Conclusion

The Secure Financial Multi-Branch Banking Hub project successfully implemented a secure WAN banking infrastructure using Cisco Packet Tracer.

The project achieved reliable communication between all branches while implementing multiple networking technologies including:

- VLSM
- VLANs
- DHCP
- DNS
- Static Routing
- NAT/PAT
- Static NAT

- Web Server Hosting
- IPsec VPN
- ICMP Testing

All required services were tested successfully, and the network achieved secure communication between the HQ, branches, and the central bank router.

The project demonstrates practical implementation of enterprise-level networking concepts and secure WAN communication.

Final Submission Checklist

Item	Status
Packet Tracer File (.pkt)	Completed
Report Documentation	Completed
Configurations Included	Completed
Screenshots Included	Completed
Connectivity Tested	Completed
VPN Verified	Completed
NAT/PAT Verified	Completed
Web Server Working	Completed
DNS Working	Completed

Appendix

Important Verification Commands

Router Commands

show ip interface brief

show ip route

show ip nat translations

show ip nat statistics

show crypto isakmp sa

show crypto ipsec sa

show ip dhcp pool

Switch Commands

show vlan brief

show interfaces trunk

Connectivity Testing

ping [destination IP]
